Lecture 10: The Physics of Neutron Moderation

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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1 Introduction: The Fork in the Road (Fast vs. Thermal)

In Lecture 9, we defined the cross section (σ). In this lecture we shall show that for Uranium-235, the probability of fission is massive at low energies ($\sim$580 barns at 0.025 eV) but tiny at high energies ($\sim$1 barn at 2 MeV).

When a fission event occurs, the new neutrons are born **Fast** ($E \approx 2$ MeV). This presents the nuclear engineer with two fundamental choices:

1.1 Option A: The Thermal Reactor (The focus of this course)

We deliberately slow the neutrons down using a **Moderator** (water, graphite) to take advantage of the massive low-energy cross section.

- **Pros:** We can use natural or slightly enriched uranium (3-5%). The neutron economy and reactivity feedbacks are easier to control.
- **Cons:** We lose neutrons to absorption during the slowing-down process.
- **Examples:** LWR (USA), CANDU (Canada), HTGR.

1.2 Option B: The Fast Reactor

We keep the neutrons moving at high speeds (no moderator) to fission ^{238}U or breed Plutonium.

- **Pros:** At high energies, the neutron yield per fission (ν) is higher, allowing for "Breeding" (creating more fuel than is consumed). This allows us to close the fuel cycle (as done in France/Russia).
- **Cons:** Since the cross section is so small ($\sim$1 barn), typically requires higher fissile content (often ¿20% U-235 or plutonium-bearing fuel) to sustain a chain reaction. The coolant cannot contain hydrogen (usually liquid Sodium or Lead).

Decision: For the next several weeks, we will focus exclusively on **Option A**. To understand Thermal Reactors, we must understand the mechanics of how a neutron loses energy. This process is called **Moderation**.

2 The Life Cycle of a Neutron (Part 1)

In Lecture 9, we defined the probability of interaction (the cross section, σ). Today, we look at the consequences of those interactions. Our goal in a thermal nuclear reactor is to take a "fast" neutron (born at high energy from fission) and slow it down to "thermal" energies where it can easily cause more fission.

This involves distinct physics problems:

1. **Moderation:** How does a neutron lose energy when it hits a nucleus? (Scattering mechanics).
2. **Probability of Scattering:** How likely is a collision at different energies?
3. **Absorption (Losses):** How do we avoid the "traps" (resonances) while slowing down?

3 Dynamics of Neutron Scattering (Moderation)

Primary Reference: Lamarsh, 4th Ed., Section 3.5

To achieve the "Thermal Reactor" described in Option A, we must reduce the neutron energy from 2 MeV to 0.025 eV. This is a factor of 10^8 reduction.

3.1 Energy Loss in a Single Elastic Collision

Consider a neutron (mass 1) hitting a target nucleus (mass A) in an elastic "billiard ball" collision. Conservation of momentum and energy dictates how much energy is transferred.

Visualizing the Physics: The "Head-On" Collision ($A = 1$)

The mechanics of moderation are best understood by looking at the specific case of a neutron hitting a Hydrogen nucleus ($A = 1$) head-on. The easiest way to solve this is to shift our perspective to the **Center of Mass (COM)** frame of reference.

1. **Lab Frame (Before):** The neutron moves at velocity v. The proton is stationary. The "average" velocity of the system (Center of Mass) is $v_{cm} = v/2$.
2. **COM Frame:** We move along with the center of mass.
   - The neutron appears to move right at $v - v/2 = +v/2$.
   - The proton appears to move left at $0 - v/2 = -v/2$.
3. **The Collision:** Since momentum must be zero in the COM frame and energy is conserved, the particles simply bounce off with the same speed but opposite direction.
   - Neutron rebound velocity (COM) $= -v/2$.
4. **Lab Frame (After):** We shift back to the lab frame by adding v_{cm} to the result.
   - Neutron Final Velocity $= (-v/2)_{rebound} + (+v/2)_{frame} = \mathbf{0}$.

Conclusion: In a head-on collision with Hydrogen, the neutron transfers **100%** of its momentum and energy to the proton, stopping dead in its tracks.

Generalizing to Mass A

For heavier nuclei ($A > 1$), the center of mass moves much slower than $v/2$, so the neutron can never lose all its energy. The maximum possible energy loss is determined by the **Collision Parameter**, α:

$$\alpha = \left(\frac{A-1}{A+1}\right)^2 \tag{1}$$

The neutron's energy after one collision will fall somewhere in the range:

$$\alpha E_{in} \leq E' \leq E_{in} \tag{2}$$

Note:

- **Hydrogen ($A = 1$):** $\alpha = 0$. The neutron can lose up to 100% of its energy (as derived above).
- **Lead ($A = 208$):** $\alpha \approx 0.98$. The neutron retains at least 98% of its energy even in a perfect head-on collision.

3.2 Average Logarithmic Energy Decrement (ξ)

Since neutrons scatter at random angles, we care about the *average* behavior. We define ξ (xi) as the average loss in the natural log of energy per collision:

$$\xi = \ln\left(\frac{E_{in}}{E_{out}}\right)_{average} \approx \frac{2}{A + 2/3} \quad \text{(for } A > 1\text{)} \tag{3}$$

This parameter dictates our choice of moderator:

- **Water (H):** $\xi = 1.0$. Requires $\sim$18 collisions to thermalize.
- **Graphite (C):** $\xi = 0.158$. Requires $\sim$115 collisions.
- **Uranium (U):** $\xi = 0.008$. Requires $\sim$2170 collisions.

3.3 Energy Dependence of Elastic Scattering

So far, we have discussed *how much* energy is lost if a collision occurs. But how *likely* is the collision?

For most good moderators (like Hydrogen, Carbon, and Oxygen) in the energy range of interest (1 eV to 100 keV), the elastic scattering cross-section (σ_s) is relatively constant.

- This is often called **Potential Scattering**.
- The neutron essentially sees the nucleus as a hard sphere with cross-section $\sigma_s \approx 4\pi R^2$, where R is the nuclear radius.
- **Significance:** Unlike absorption (which varies wildly, see Section 4), the scattering probability is reliable. This creates a steady "staircase" of energy loss, allowing the neutron to reliably march down from MeV energies toward thermal energies.

3.4 Inelastic Scattering: The "Heavy Lifter" at High Energy

You might ask: "Doesn't inelastic scattering lose even more energy?" Yes. In an inelastic collision ($n + A \to n' + A^*$), the neutron gives up a massive amount of kinetic energy to excite the nucleus.

- **Significance:** This mechanism is very effective for **Heavy Nuclei** (like Uranium or Iron) at high energies (> 1 MeV).
- **The Limitation:** It is a **Threshold Reaction**. It can only occur if the neutron has enough energy to reach the first excited state of the nucleus.
  - For U-238, the threshold is low (~ 45 keV).
  - For Oxygen-16 (moderator), the threshold is huge (~ 6 MeV).

Conclusion: Since fission neutrons are born at ~ 2 MeV, they *cannot* inelastically scatter against the Oxygen in the water. We must rely on **Elastic Scattering** to take the neutron the rest of the way down.

3.5 The Engineer's Trade-off: Moderating Ratio vs. Economics

To select a moderator, a nuclear engineer must balance the "slowing down power" against the "absorption penalty." We define the **Moderating Ratio (MR)**:

$$MR = \frac{\xi \Sigma_s}{\Sigma_a} \tag{4}$$

While the Moderating Ratio ($MR = \xi\Sigma_s/\Sigma_a$) tells us which material is the most efficient purely in terms of neutron economy, it ignores two massive engineering constraints: **Cost** and **Core Size**.

- **The Physics of Size:** The scattering cross-section of Deuterium ($\sigma_s \approx 7$ b) is significantly lower than Hydrogen ($\sigma_s \approx 20$ b). This means neutrons in D_2O travel much further between collisions. To prevent leakage, a Heavy Water reactor must be physically larger than a Light Water reactor.
- **The Economics:** D_2O requires expensive isotopic separation ($\$300/kg$), whereas H_2O is effectively free.

Moderator	**MR**	**Core Size**	**Economic Trade-off**
Light Water (H_2O)	71	Small	**High Fuel Cost.** (Requires Enrichment). **Low Capital Cost.** (Compact vessel, cheap moderator).
Heavy Water (D_2O)	5670	Large	**Low Fuel Cost.** (Natural Uranium). **High Capital Cost.** (Expensive moderator, large vessel).
Graphite (C)	192	Massive	**Very Large Vessel.** (Low density, low σ_s). Cheap moderator, but expensive construction.

Historical Note: The "Impurity" that Changed the War
In 1941, German scientists tested graphite as a moderator but abandoned it for Heavy Water (D_2O) after measurements showed it absorbed too many neutrons. This failure stalled their program.

The Mistake: They used industrial graphite contaminated with trace amounts of **Boron**.

- **The Culprit:** Boron-10 ($\sigma_a \approx 3840$ barns at thermal energy).
- **The Physics:** Boron follows the **1/v Law**. It is harmless to fast neutrons but becomes a "black hole" for the thermal neutrons needed to sustain the chain reaction.

Leo Szilard realized that even ppm levels of Boron would poison the pile. He demanded "Nuclear Grade" graphite (free of Boron), allowing Fermi to build the Chicago Pile-1 successfully using cheap graphite instead of rare Heavy Water.

4 Neutron Absorption (Losses)

Primary Reference: Lamarsh Section 3.6

Now that we know *how* the neutron slows down (Section 3), we must examine the treacherous path it must travel. While we want the neutron to scatter, we desperately want to avoid it being **Captured** (Absorbed) before it reaches thermal energy.

4.1 The "Valley of Death" (Resonance Region)

Between 1 eV and 10 keV, the capture cross-section of ^{238}U exhibits a "forest" of sharp, jagged peaks.

- **The Physics:** These peaks correspond to discrete quantum energy levels of the excited compound nucleus (^{239}U*).
- **The Trap:** In ^{238}U, these are almost exclusively **Capture Resonances** (n, γ). If a neutron slows down and lands in one of these energy "traps" (e.g., the massive resonance at 6.67 eV), it is absorbed. It dies without causing fission.
- **The Name:** We call this the "Valley of Death" because neutrons must successfully "jump" over these peaks via elastic scattering to reach the safe thermal region.

4.2 Visualizing the Resonances (Interactive Tool)

Since we cannot see this behavior in a static textbook plot, we will use an online nuclear data plotter. *Note: The standard NNDC site is often offline.*

Primary Tool: KAERI Table of Nuclides `https://atom.kaeri.re.kr/nuchart/`

Exercise for Students:

1. Go to the URL above.
2. In the "Nuclide" search box, type: `U238` (and hit Enter).

3. **Choose the Library:** Locate the column labeled **ENDF/B-VIII.0** (this is the US standard) and click on the small **+** sign to expand the options.
4. **Select the Data:** In the expanded list, locate the row labeled **Capture cross sections** and click the button labeled **Plot**.
5. **Verify Scale:** The graph that appears is the (n, γ) radiative capture cross section. The axes should default to **Log-Log** scale. (If they appear linear, use the settings to switch them to Log).
6. **Observation:** Look at the energy range between 10^0 eV and 10^4 eV. You will see the massive peaks rising 1000x above the baseline. These are the "Resonance Traps" our moderator design must defeat.

4.3 Region 3: The 1/v Region (Thermal)

Once the neutron passes the last resonance (at roughly 6 eV), it enters the "safe" thermal region. Here, the absorption cross-section rises smoothly according to the **1/v Law**:

$$\sigma_a \propto \frac{1}{v} \propto \frac{1}{\sqrt{E}} \tag{5}$$

This massive rise at low energy is what makes thermal reactors possible.

Physical Interpretation: Concentration vs. Flux

A powerful way to understand low-energy neutron interactions is to distinguish between the nature of **capture** (absorption) and **scattering**.

1. Capture as a Chemical Reaction (Concentration Dependent)

At low energies, neutron capture behaves less like a projectile hitting a target and more like a second-order chemical reaction. The reaction rate depends on the local **concentration** of neutrons (n), rather than the flux. The neutron is interacting with a field rather than a hard shell; the slower it moves, the longer it lingers near the nucleus, and the higher the probability of interaction.

Mathematically, if the Reaction Rate (R) is proportional only to the product of neutron density (n) and target density (N):

$$R \propto nN \tag{6}$$

However, the standard nuclear engineering definition of reaction rate is $R = \Sigma\phi = N\sigma_a(nv)$. For these two definitions to agree, the product $\sigma_a v$ must be constant. Therefore, the effective cross-section must vary inversely with velocity:

$$\sigma_a \propto \frac{1}{v} \tag{7}$$

In this view, the "cross-section" for absorption is not a fixed geometric size, but a dynamic value that adjusts for velocity to keep the reaction rate density-dependent.

2. Scattering as a Geometric Collision (Flux Dependent)

In contrast, low-energy elastic scattering behaves like a classic "hard sphere" collision. The nucleus presents a fixed geometric target (roughly $4\pi R^2$) to the neutron. Here, the reaction rate *does* depend on the flux ($\phi = nv$): the faster the neutrons move, the more frequently they strike the target.

$$R_{scatter} \propto \phi \tag{8}$$

Because the rate scales with v, the scattering cross-section σ_s remains **constant** (flat) in the thermal region, unlike the absorption cross-section.

The Anomaly at Very Low Energy ($E \ll kT$)

There is an important exception to the rule that elastic scattering is constant. If the neutron energy drops **significantly below** the thermal energy of the moderator (e.g., "Cold Neutrons" in the meV range), the elastic scattering cross-section appears to rise again as $1/v$.

This is not due to a change in the nuclear force, but rather the **thermal motion of the target**.

Standard reactor physics definitions assume the target nucleus is at rest. However, when $v_{neutron} \ll v_{target}$ (where v_{target} is the thermal vibration speed of the lattice), the collision rate is determined by the target atoms hitting the effectively stationary neutron.

The physical reaction rate becomes proportional to the target's thermal speed v_{th} rather than the neutron's speed:

$$R \propto v_{th} \quad \text{(constant)} \tag{9}$$

Since we define cross-section $\sigma(v)$ such that $R = nv\sigma(v)$, we find:

$$nv\sigma(v) \propto \text{constant} \implies \sigma(v) \propto \frac{1}{v} \tag{10}$$

Thus, at extremely low energies, the elastic scattering cross-section mimics the $1/v$ behavior of absorption because the interaction is driven by the "flux" of the target atoms, not the neutron.

What is "Thermal" Energy?

When we say the neutron slows to "Thermal" energy, we mean it has reached thermal equilibrium with the moderator medium (the water molecules). The neutrons are no longer losing energy on average; they are bouncing around, gaining and losing small amounts of energy from the vibrating water molecules.

The neutron energies follow a **Maxwell-Boltzmann Distribution**, effectively behaving like a gas at temperature T:

$$E_{mp} \sim k_B T \tag{11}$$

At room temperature (293 K), the most probable energy is 0.0253 eV (velocity $\approx$ 2200 m/s). In a hot reactor (600 K), the spectrum shifts to higher energies ("Harder Spectrum"), which affects how the reactor operates.

5 Summary: The Goal of Lecture 10

We have successfully described the journey of the neutron from birth (high energy) to thermal maturity (low energy).

- We defined the **Moderator** (H, C) and why we use it (ξ).
- We analyzed the **Mechanics** of energy loss (α, COM frame).
- We identified the **Enemy**: Resonance Capture in U-238.

However, we have not yet discussed the most important event: **Death and Rebirth**. What happens when the thermal neutron finally hits a ^{235}U nucleus?

In **Lecture 11 (Friday)**, we will cover the **Physics of Fission**, the Liquid Drop Model, and the origin of the massive energy release that powers the reactor.